

Starting smartly with charts

DATA STORYTELLING CONCEPTS



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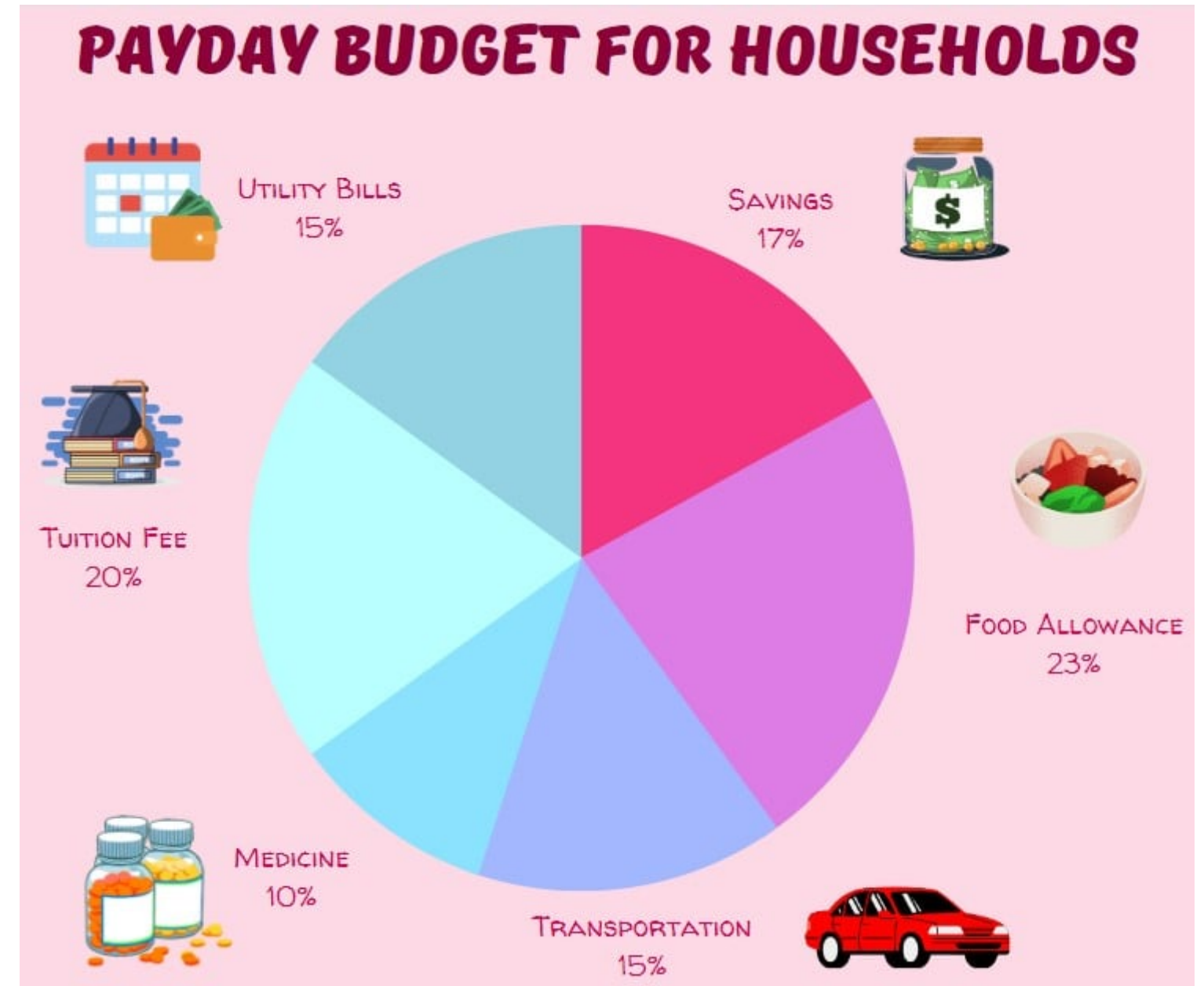
Match your data to a chart

- Certain chart types work best with certain data types
- Charts are the most common visualizations
- Charts display quantitative data in an intuitive format



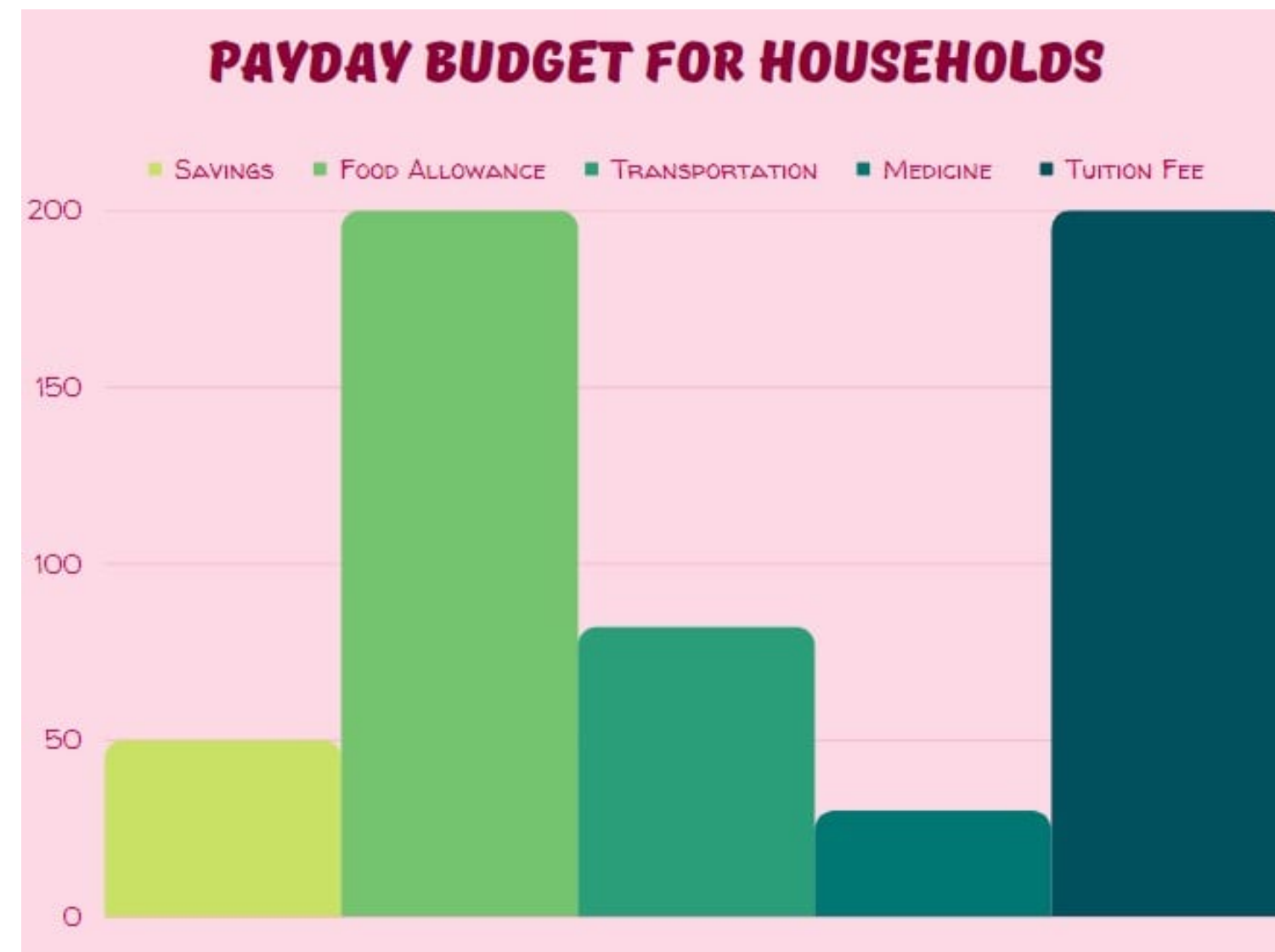
Pie charts

- Pie charts are very popular
- However, they do a poor job of comparing values with similar amounts
- Use these sparingly



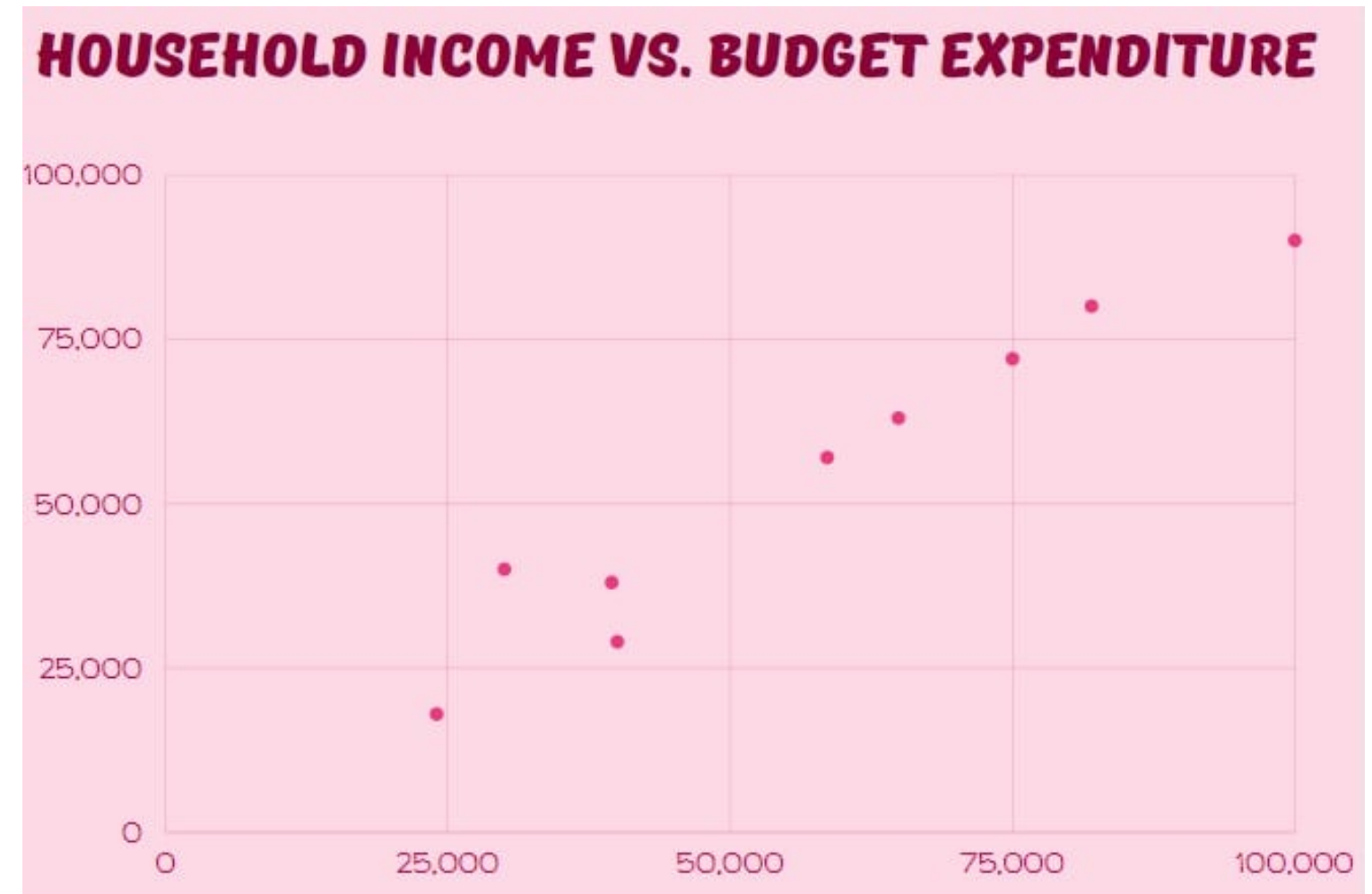
Bar charts

- Show counts and frequency by category
- When a pie chart is used, try a bar chart instead
- Bar charts require less effort from your audience



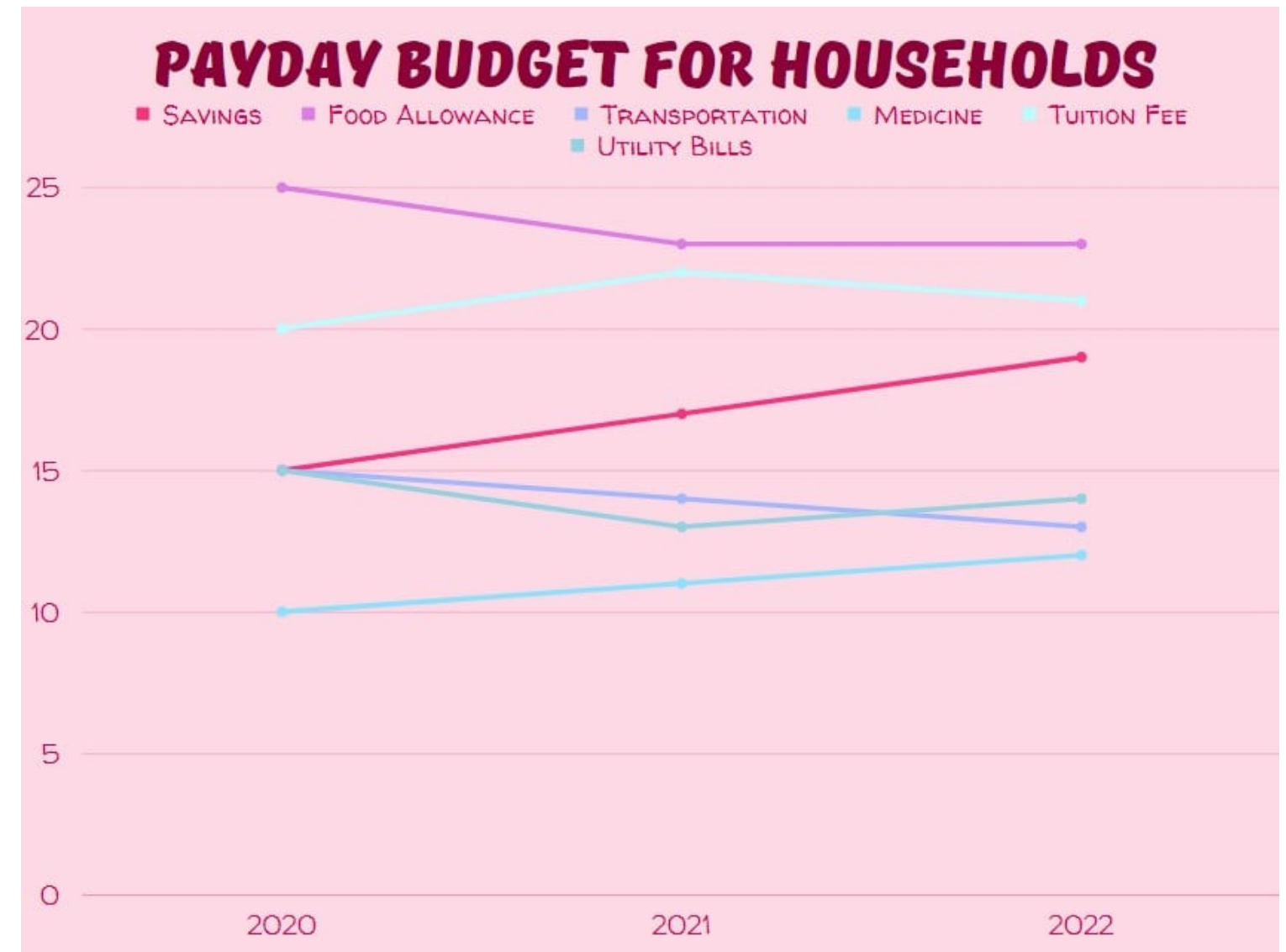
Scatter plot

- Scatter plots best represent the relationship between two quantitative variables
- Plot the observations using an X-axis and Y-axis
- Scatter plots show if the relationship between two variables is positive, negative, or relatively neutral.



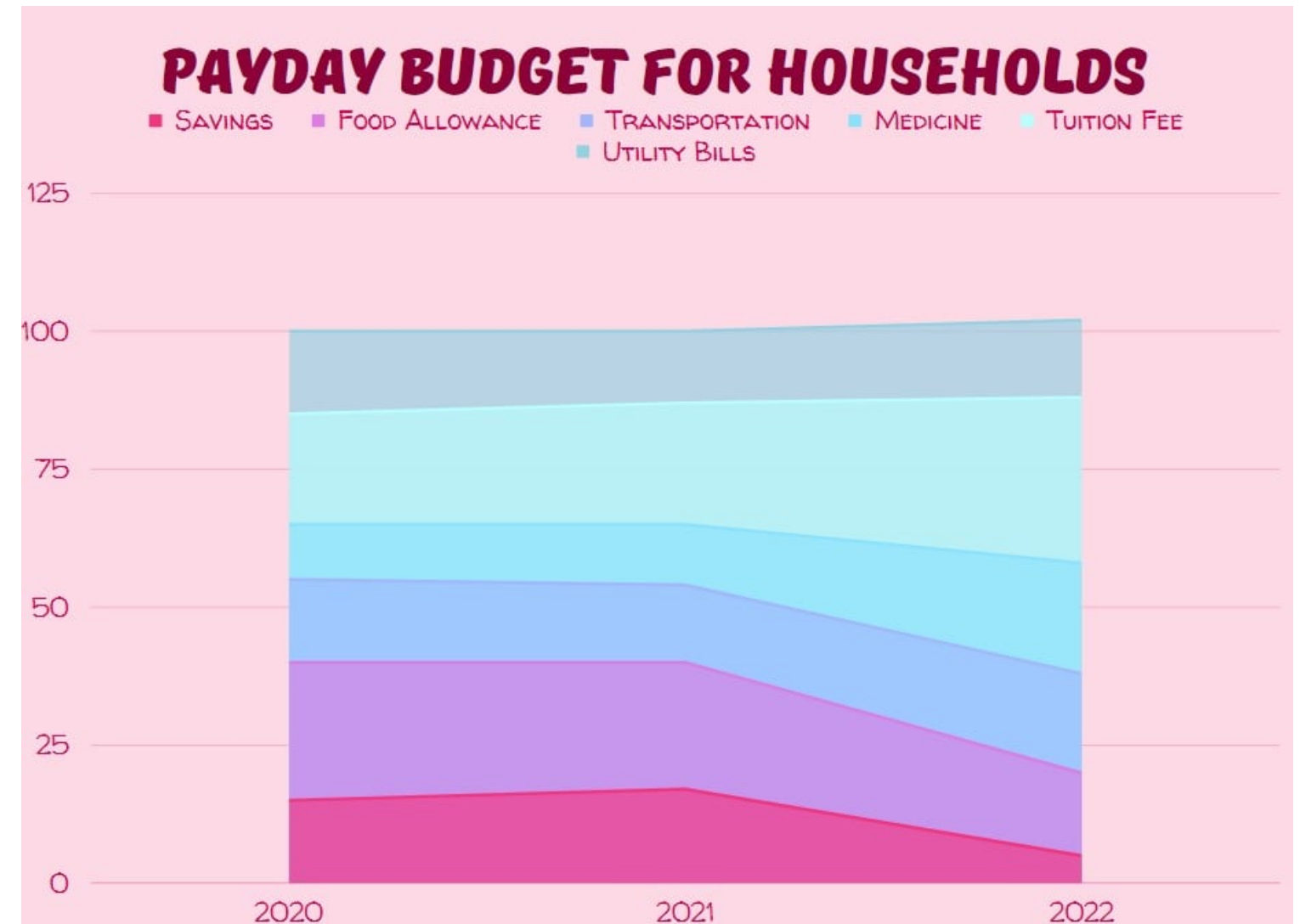
Line chart

- Line charts work well with time series data
- Compare and contrast performance over time
- Trends become easy to spot and analyze



Area charts

- Area charts show many variables over time by filling in space to the X-axis
- The areas under a trend line are different colors for different categories
- Area charts can work well when displaying a relationship between the two variables



Let's practice!
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Credibility in visualizations

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Strive to communicate

- Trust between the data storyteller and the audience is a key
- Informing someone is merely reporting something in a clinical, passive manner
- Proper communication means the audience has complete comprehension of the message
- Dual coding is an effective way to communicate



Best practices

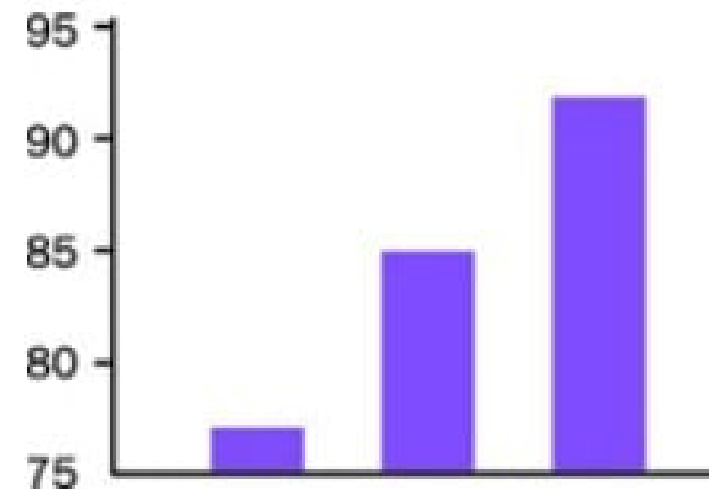
- Consistency and a reputation for integrity are important to practice
- Consider having a subject matter experts speak about specific visuals
- Cite your information from credible sources



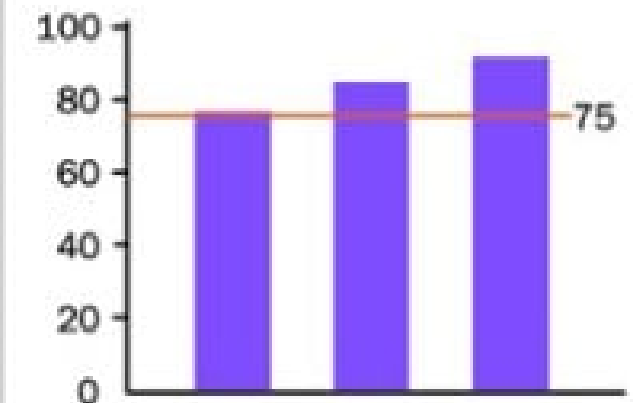
Truncation without causation

- Avoid truncating your Y-axis
- This negates the advantages of a bar chart
- If you must truncate, make a note on the document

NO



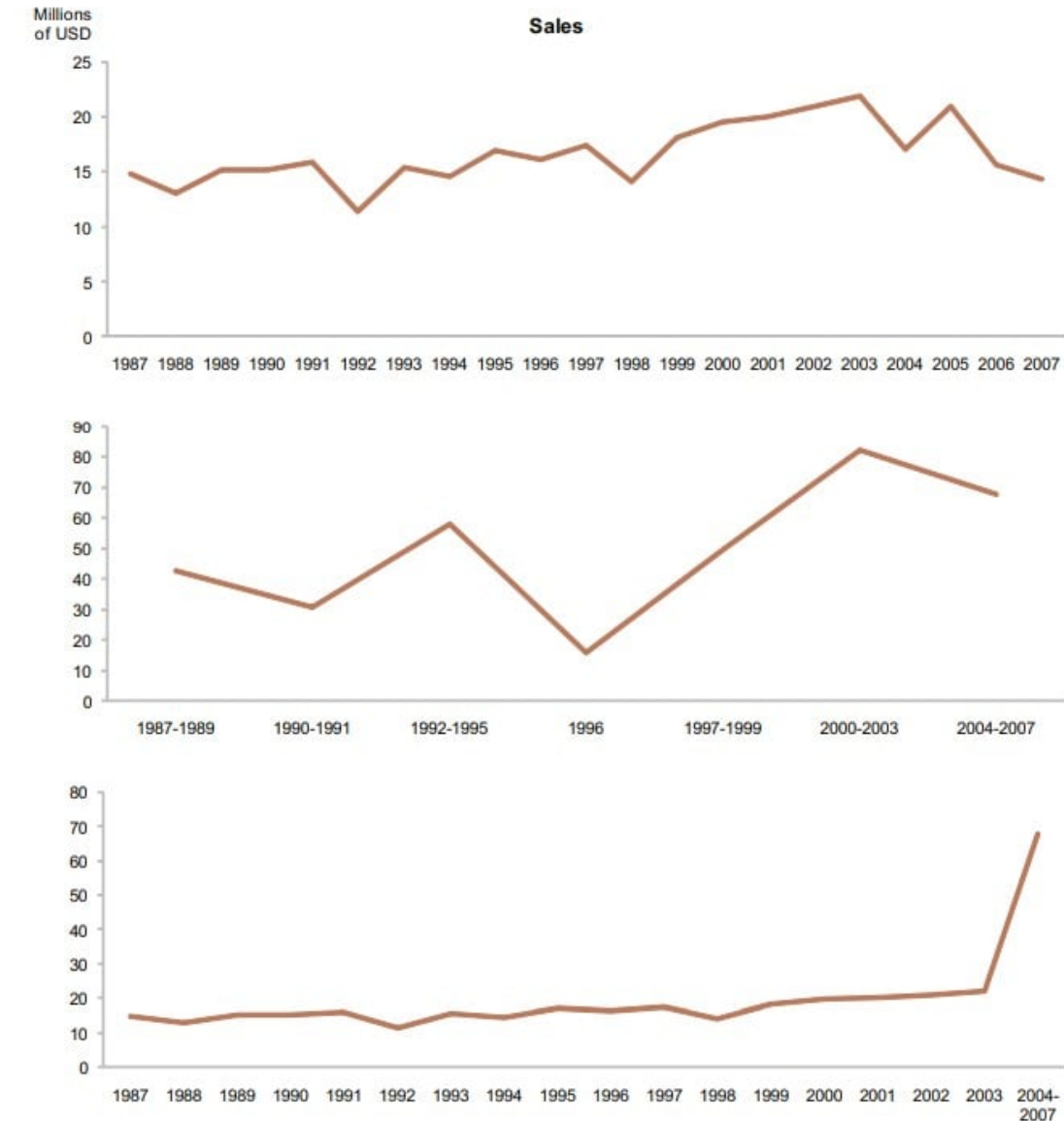
YES



¹ Dykes, Brent. "Data Storytelling: How to Drive Change with Data, Narrative and Visuals," Wiley. 2020

At an appropriate interval

- Inconsistent data intervals are to be avoided
- Within a single axis, each interval should represent the same value
- Simple charts are easy to understand charts and thus easier to trust
- Only the first chart on the top shows the X-axis with consistent intervals



¹ Few, Stephen. "Line Graphs and Irregular Intervals: An Incompatible Partnership"

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Eliminate clutter and focus attention

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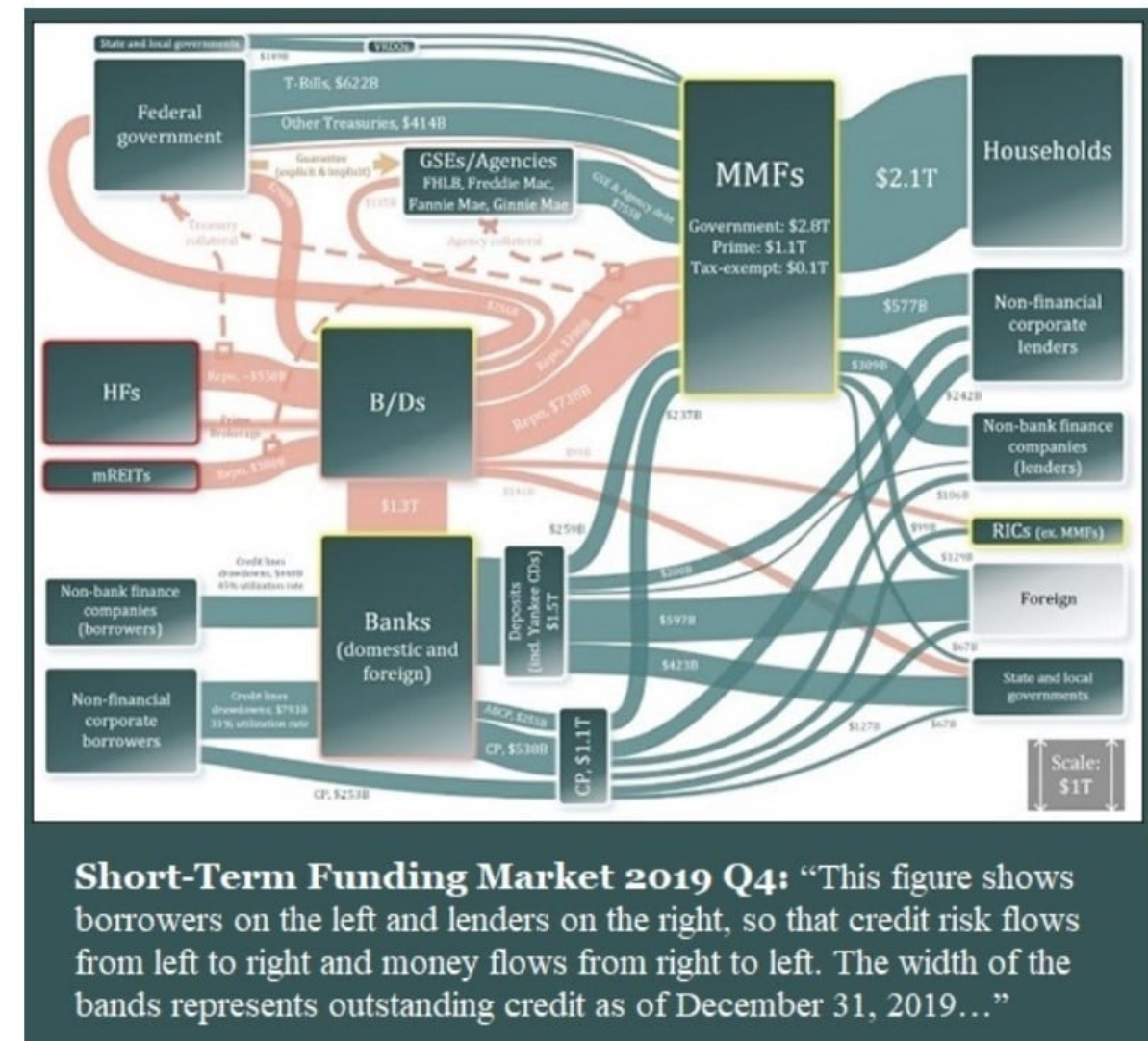


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Cluttered charts are counter-productive

- More design elements does not mean "better"
- Clutter distracts and promotes miscommunication
- Absorb your audience in the *problem* without unnecessary effort on their part



¹ https://www.sec.gov/files/US-Credit-Markets_COVID-19_Report.pdf

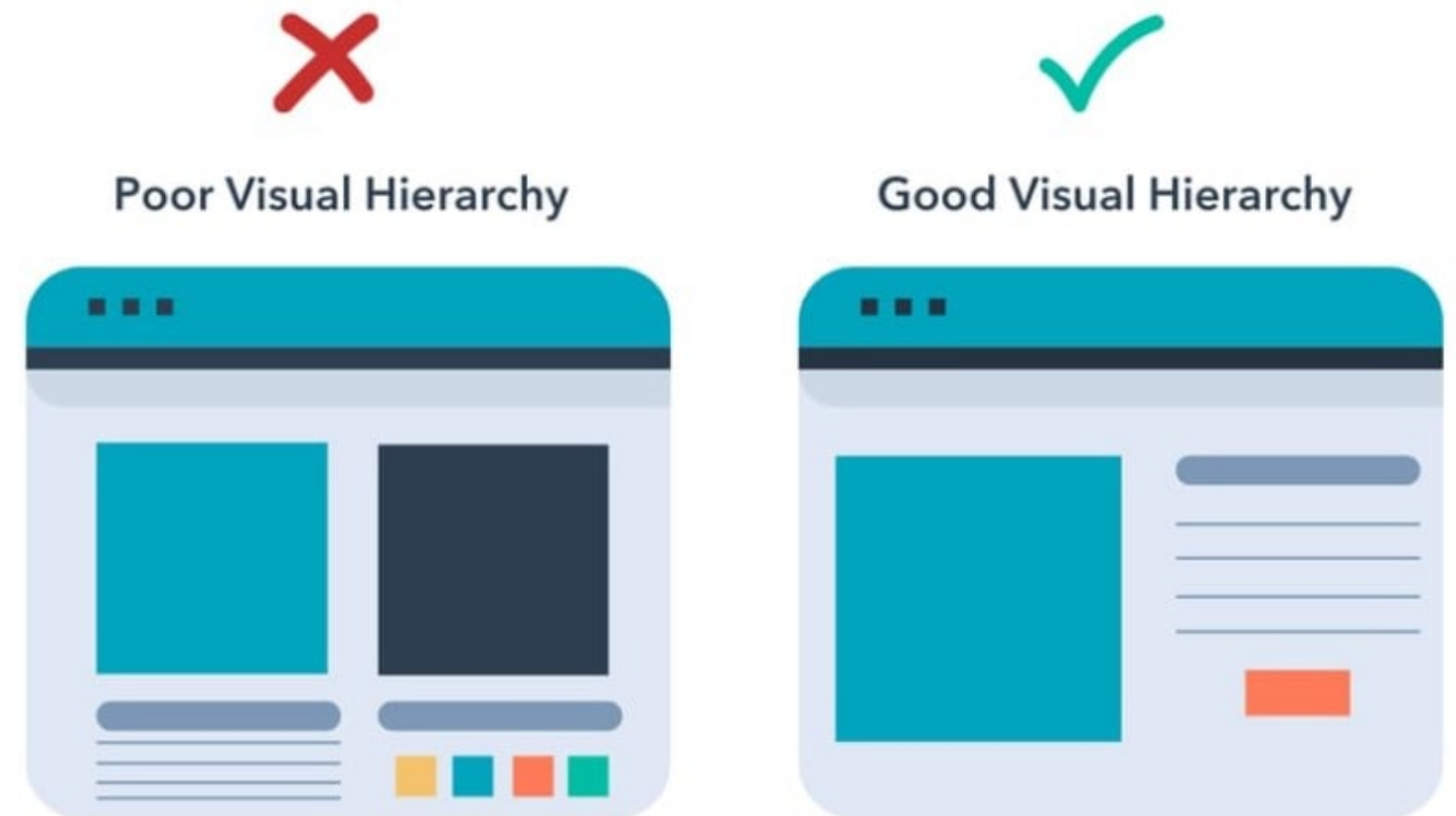
Sensory overload

- Sometimes you want to share aggregate data
- Tables can help but should be used sparingly and almost never on slides
- Get started by making a "bad" chart and refine that with help as necessary

Exp. 1	1		2		3		4		5		6		7		8		9		10		Total	
Label	Total		Total		Total		Total		Total		Total		Total		Total		Total		Total		Total	
	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No
A	29	0	29	1	30	0	26	0	29	0	28	0	30	0	27	0	30	0	28	0	286	1
B	7	2	8	7	9	2	8	4	8	4	6	2	6	2	7	3	9	5	10	4	78	35
C	7	0	7	0	10	0	12	0	8	1	8	0	11	0	13	0	10	0	11	0	97	1
D	10	0	7	0	8	0	9	0	6	0	9	0	7	0	7	0	9	0	10	0	82	0
E	4	0	4	0	7	0	3	0	3	0	5	0	5	0	6	0	3	0	8	0	48	0
F	6	2	6	5	6	3	7	4	7	4	5	4	6	5	8	3	5	4	7	3	63	37
G	4	0	4	0	4	0	4	0	4	0	4	0	4	0	4	0	4	0	4	0	40	0
H	5	1	5	1	5	1	5	1	5	2	5	1	5	2	5	2	5	2	5	2	50	15
Total	72	5	70	14	79	6	74	9	70	11	70	7	74	9	77	8	75	11	83	9	744	89
%	94%	6%	83%	17%	93%	7%	89%	11%	86%	14%	91%	9%	89%	11%	91%	9%	87%	13%	90%	10%	89%	11%

Can I have your attention please?

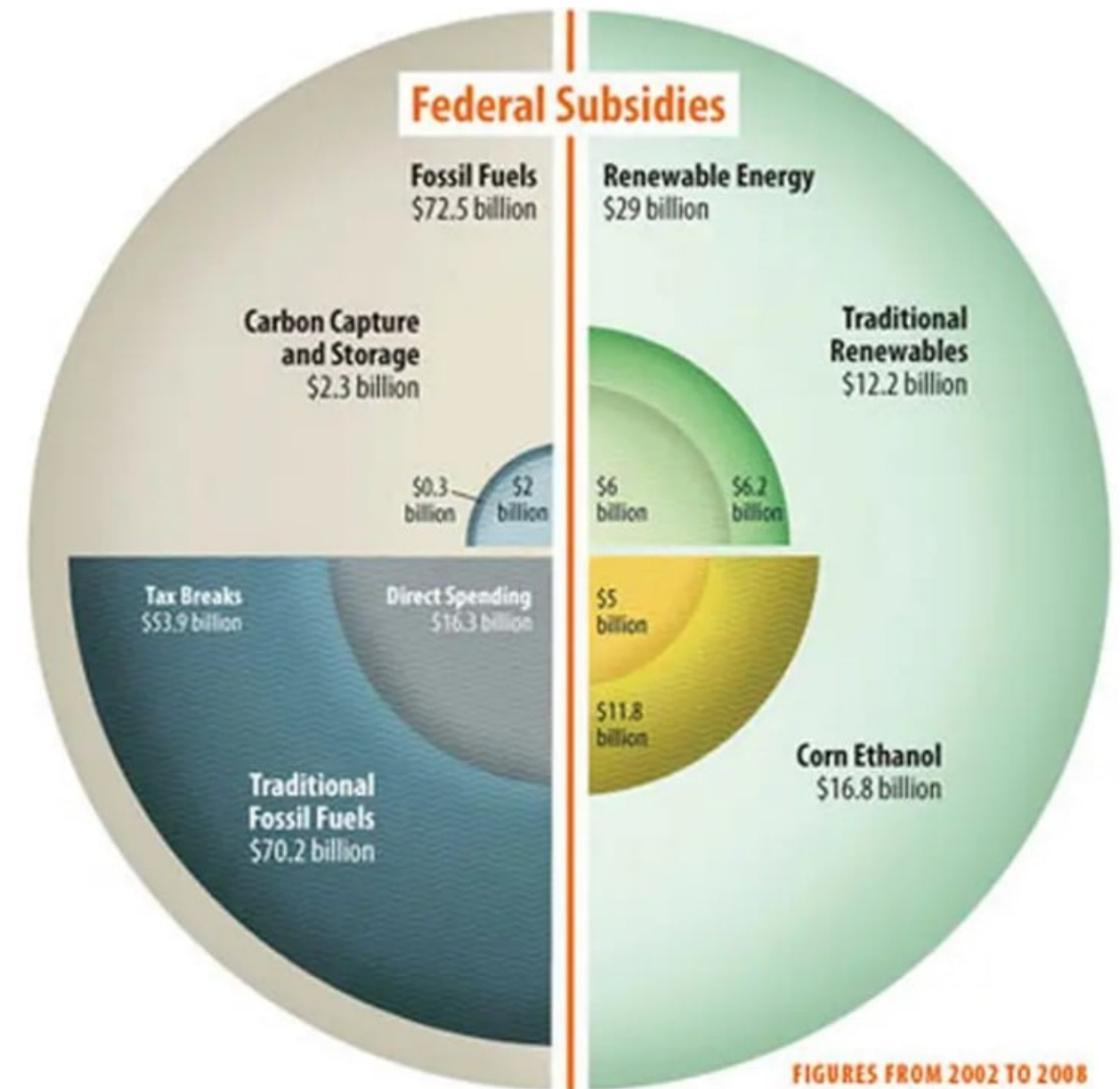
- A visual hierarchy signals where to look
- Size, hue, and spatial position signal relative importance
- People follow charts the way that they read documents; for English, left to right



¹ <https://blog.hubspot.com/marketing/visual-hierarchy>

I just don't follow you

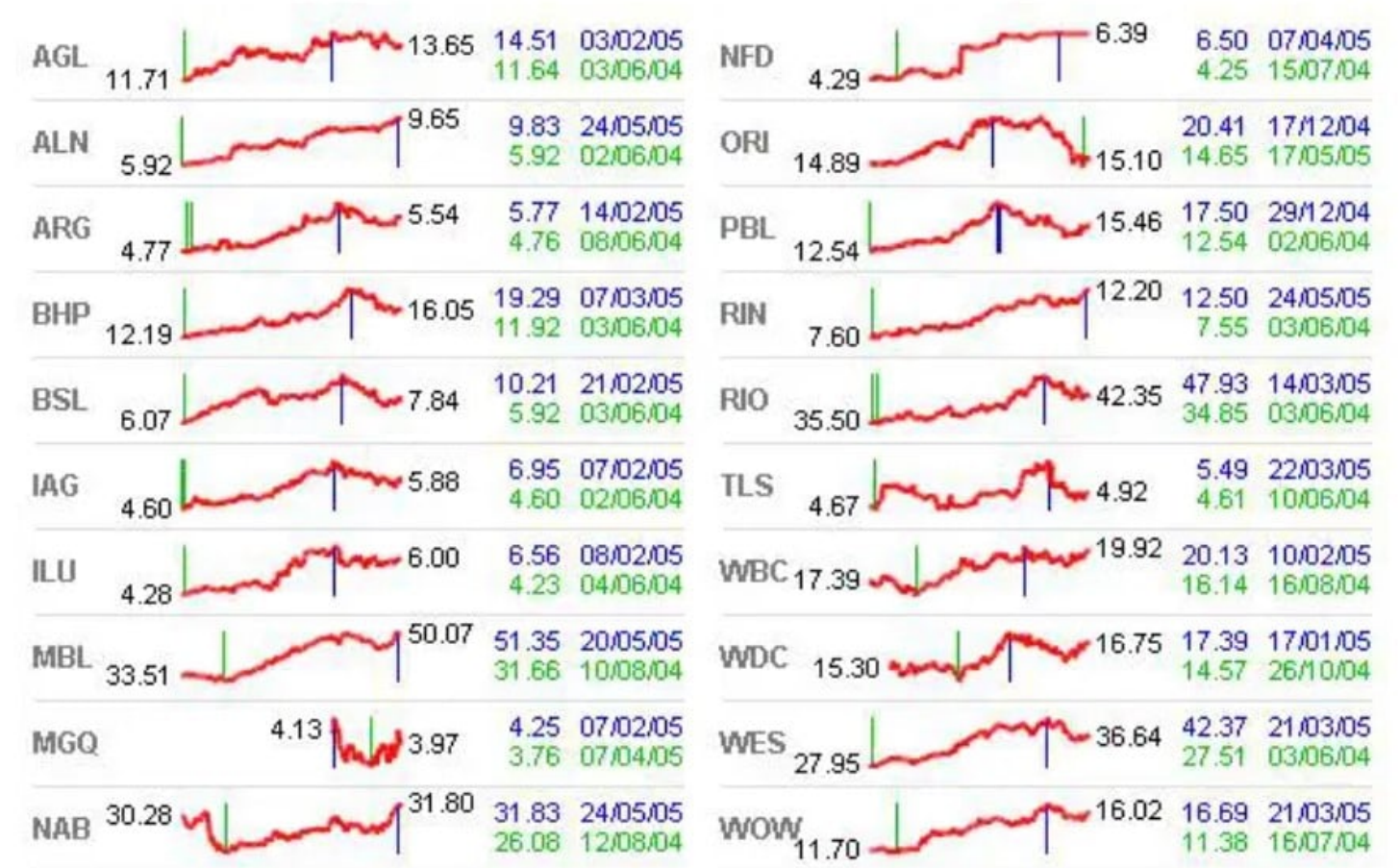
- Don't include unnecessary information or design elements
- Pointless objects invite distraction
- Just because a visualization is interesting does not mean it is helpful
- Information can be split up into more than one chart



¹ http://junkcharts.typepad.com/junk_charts/2011/04/worst-statistical-graphic-nominated.html

Data density

- Data density is the ratio of the information displayed to how much space it occupies
- The ideal data density in a visualization is around 50%-60%
- Proper data density makes visual details memorable
- Higher density charts communicate a lot of information but are not always appropriate



¹ edwardtufte.com

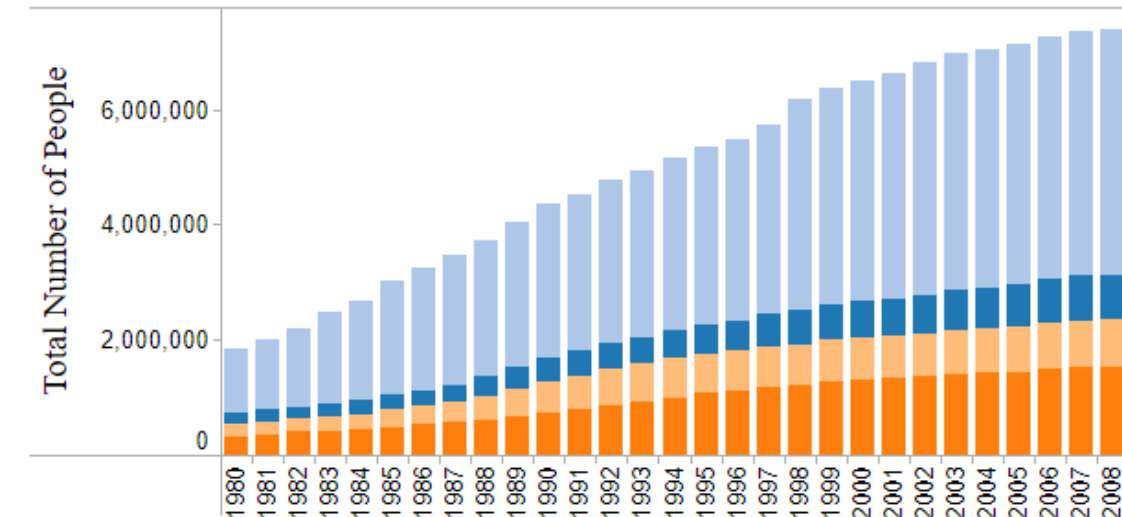
Never just casually colorful

- Color should emphasize only specific insights
- Multiple colors are for highlighting different trends
- Orange and blue are a color-blind friendly palette

A Look at the U.S. Correctional Population

In 2008, 1 in every 31 adults or 3.2 percent of all U.S. adults were in the system.

Data Source: Bureau of Justice Statistics



2008 Correctional Population: 7,308,200

The number of people locked up in prison or in jail more than **tripled** since 1980, according to BJS data. The number on probation was up by **282%** and the number on parole was up over **275%** since 1980.

Census data show that the general population grew by **one-third** over the same time period from 227 million in 1980 to 304 million in 2008.

On Probation
In Jail
On Parole
In Prison

Compare Sentences Below

Use the categories below to compare four levels of correctional supervision.

☒ (All)
☒ In Jail
☒ In Prison

On the Web
stateoftheusa.org

On Twitter
@stateoftheusa

On Facebook
facebook.com/

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Storyboard your results

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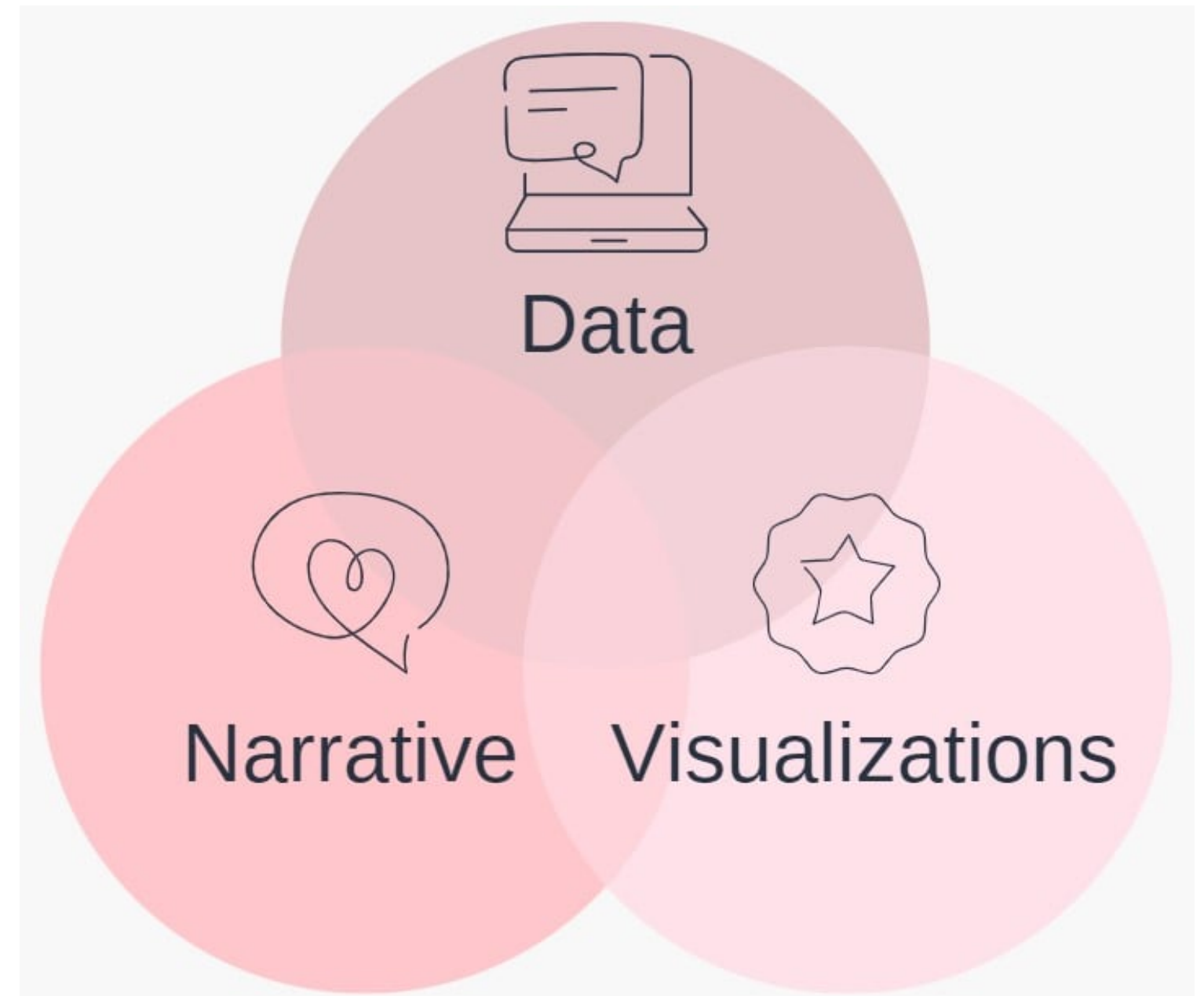


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Why do a storyboard?

- If you have a written narrative, it is still beneficial to storyboard
- Data stories blend data, a narrative, and visuals
- Narrative and storyboards both tell a meaningful and memorable story and inspire change



Ask me how do I feel

Your storyboard is your graphic organizer

- Ask yourself how the visuals make you feel
- Ask yourself if you are better informed after looking at the visuals
- Ask yourself what you want your audience to think and feel



Storyboard construction

- Sort your visuals in terms of beginning, middle, and end
- There is one challenge and the main message
- Feel free to explore different timelines
- The final result should be a clear, cohesive story and message



Cue the theme music

- Example themes
 - Sales must improve soon
 - Better hiring and retention practices are needed
 - Our rebranding efforts have succeeded beyond our expectations
- Describing your visuals aloud feels silly but is actually very smart



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Wrap-up

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Recap

- **You learned about:**
 - Why data stories are useful
 - Structuring data stories
 - Authoring impactful visuals

Wide world of data storytelling waiting to be unlocked.

Additional Resources

Books

- Effective Data Storytelling: How to Drive Change with Data, Narrative and Visuals
- DataStory: Explain Data and Inspire Action Through Story
- Good Charts: The HBR Guide to Making Smarter, More Persuasive Data Visualizations
- Storytelling with Data: A Data Visualization Guide for Business Professionals
- The Big Book of Dashboards: Visualizing Your Data Using Real-World Business Scenarios

Congratulations!

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